

Exp. No. 31

Implement Lexical Analyzer using FLEX (Fast Lexical Analyzer). The program should separate the tokens in the given C program and display with appropriate caption.

Input Source Program: (sample.c)

```
#include<stdio.h>

void main()
{
    int a,b,c = 30;

    printf("hello");
}
```

Program: (token.l)

```
digit [0-9]
letter [A-Za-z]
%{
int count_id,count_key;
}%
%%
(stdio.h|conio.h) { printf("%s is a standard library\n",yytext); }
(include|void|main|printf|int) { printf("%s is a keyword\n",yytext); count_key++; }
{letter}({letter}|{digit})* { printf("%s is a identifier\n", yytext); count_id++; }
{digit}+ { printf("%s is a number\n", yytext); }
\"(\\.|[^\"\\])*\" { printf("%s is a string literal\n", yytext); }
.|\\n { }
%%
int yywrap(void) {
return 1;
}
int main(int argc, char *argv[]) {
yyin = fopen(argv[1], "r");
yylex();
printf("number of identifiers = %d\n", count_id);
printf("number of keywords = %d\n", count_key);
fclose(yyin);
}
```

Output:

```
C:\Users\Administrator\Documents\exp 31.exe

G:\lex>flex token.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe sample.c
include is a keyword
stdio.h is a standard library
void is a keyword
main is a keyword
int is a keyword
a is a identifier
b is a identifier
c is a identifier
30 is a number
printf is a keyword
"hello" is a string literal
number of identifiers = 3
number of keywords = 5

G:\lex>_
```

Exp. No. 32

Write a LEX program to count the number of vowels in the given sentence.

Program: (vowels.l)

```
%{
    int vow_count=0;
    int const_count=0;
}%

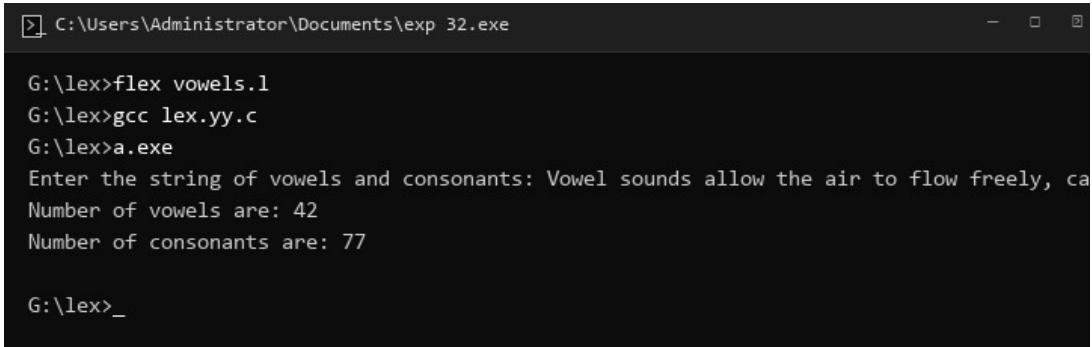
%%
[aeiouAEIOU]
{vow_count++;} [a-zA-Z]
{const_count++;}
%%
int
yywrap(){}
int main()
{
    printf("Enter the string of vowels and consonants:");
    yylex();
    printf("Number of vowels are: %d\n", vow_count);
}
```

```

    printf("Number of consonants are: %d\n", const_count);
    return 0;
}

```

Output:



```

C:\Users\Administrator\Documents\exp 32.exe
G:\lex>flex vowels.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter the string of vowels and consonants: Vowel sounds allow the air to flow freely, ca
Number of vowels are: 42
Number of consonants are: 77

G:\lex>_

```

Exp. No. 33

Write a LEX program to count the number of vowels in the given sentence.

Program: (vowels.l)

```

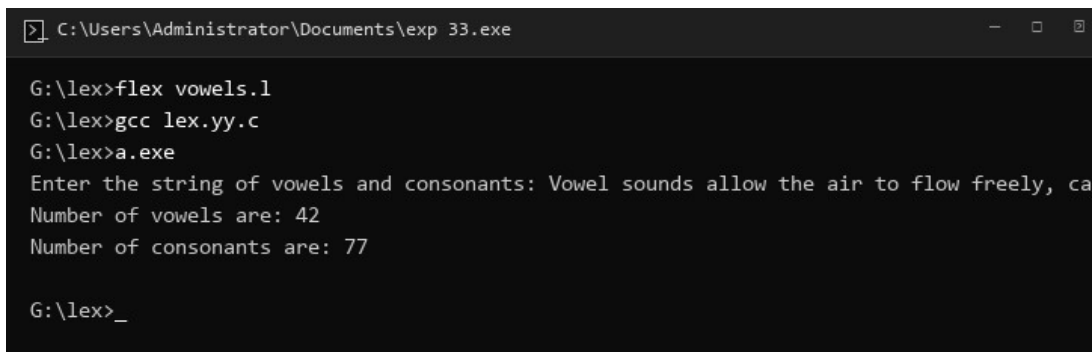
%{
    int vow_count=0;
    int const_count=0;
}%

%%
[aeiouAEIOU]
{vow_count++;} [a-zA-Z]
{const_count++;}
%%

int
yywrap(){}
int main()
{
    printf("Enter the string of vowels and consonants:");
    yylex();
    printf("Number of vowels are: %d\n", vow_count);
    printf("Number of consonants are: %d\n", const_count);
    return 0;
}

```

Output:



```
C:\Users\Administrator\Documents\exp 33.exe

G:\lex>flex vowels.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter the string of vowels and consonants: Vowel sounds allow the air to flow freely, ca
Number of vowels are: 42
Number of consonants are: 77

G:\lex>_
```

Exp. No. 34

Write a LEX program to separate the keywords and identifiers.

Input Source Program: (sample.c)

```
#include<stdio.h>
```

```
void main()
```

```
{
```

```
int a,b,c = 30;
```

```
printf("hello");
```

```
}
```

Program: (token.l)

```
digit [0-9]
```

```
letter [A-Za-z]
```

```
%{
```

```
int count_id,count_key;
```

```
%}
```

```
%%
```

```
(stdio.h|conio.h) { printf("%s is a standard library\n",yytext); }
```

```
(include|void|main|printf|int) { printf("%s is a keyword\n",yytext); count_key++; }
```

```
{letter}({letter}|{digit})* { printf("%s is a identifier\n", yytext); count_id++; }
```

```
{digit}+ { printf("%s is a number\n", yytext); }
```

```
"(\.|\.[^"\\"])*" { printf("%s is a string literal\n", yytext); }
```

```
.\n { }
```

```
%%
```

```
int yywrap(void) {
```

```
return 1;
```

```
}
```

```
int main(int argc, char *argv[]) {
```

```
yyin = fopen(argv[1], "r");
```

```
yylex();
```

```

printf("number of identifiers = %d\n", count_id);
printf("number of keywords = %d\n", count_key);
fclose(yyin);
}

```

Output:

```

C:\Users\Administrator\Documents\exp 34.exe

G:\lex>flex token.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe sample.c
include is a keyword
stdio.h is a standard library
void is a keyword
main is a keyword
int is a keyword
a is a identifier
b is a identifier
c is a identifier
30 is a number
printf is a keyword
"hello" is a string literal
number of identifiers = 3
number of keywords = 5

G:\lex>_

```

Exp. No. 35

Write a LEX program to recognise numbers and words in a statement.

Program: (numbers_words.l)

```

%%
[ \t ]+ ;
[0-9]+|[0-9]*\.[0-9]+ { printf("\n%s is NUMBER",
yytext);} #.* { printf("\n%s is COMMENT", yytext);}
[a-zA-Z]+ { printf("\n%s is WORD", yytext);}
\n { ECHO;}
%%

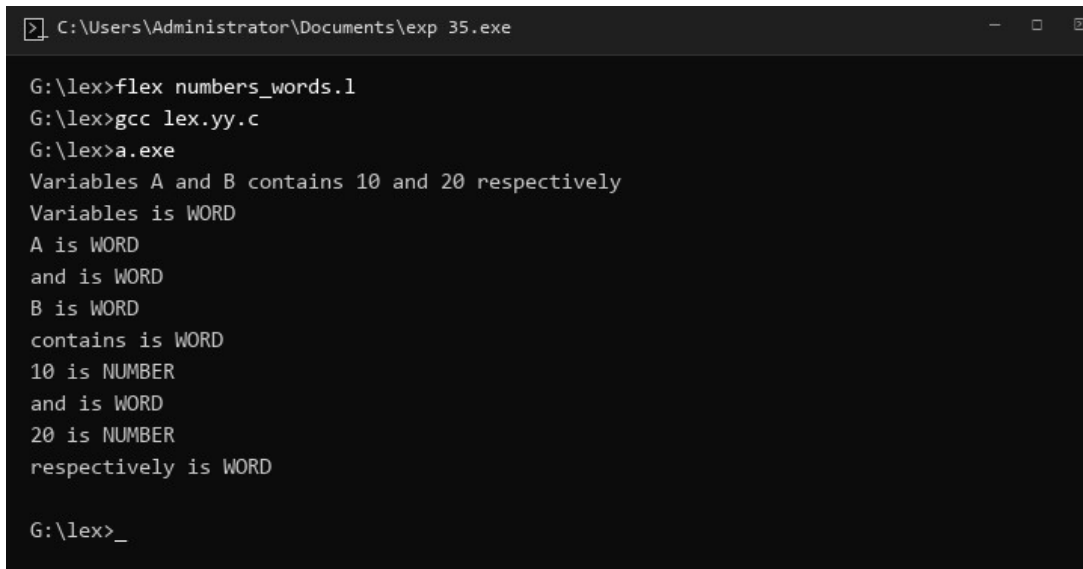
int main()
{
    while( yylex());
}

int yywrap()

```

```
{  
    return 1;  
}
```

Output:



```
C:\Users\Administrator\Documents\exp 35.exe  
G:\lex>flex numbers_words.l  
G:\lex>gcc lex.yy.c  
G:\lex>a.exe  
Variables A and B contains 10 and 20 respectively  
Variables is WORD  
A is WORD  
and is WORD  
B is WORD  
contains is WORD  
10 is NUMBER  
and is WORD  
20 is NUMBER  
respectively is WORD  
G:\lex>_
```